TIME-ITERATION METHODS FOR CONTROLLABILITY

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Abstract.

1. Linear systems of ODEs and control cost

In this section we refer to the linear time-invariant control system on $\mathbb{R}^n$ given by

$$\dot{y}(t) = Ay(t) + Bu(t) \tag{1.1}$$ {{FiniteLinear

where $y(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the control and $A \in \mathbb{R}^{n\times n}, B \in \mathbb{R}^{n\times m}$ are constant matrices.

Definition 1.1 (Controllability)**.** *Let $T > 0$. System 1.1 is said to be controllable in time T when, for any initial state $y_0 \in \mathbb{R}^n$ and any final state $y_T \in \mathbb{R}^n$, there exists $u \in L^p(0, T; \mathbb{R}^m)$ such that the solution to 1.1 with $y(0) = y_0$ satisfies $y(T) = y_T$.*

1.1. **On the cost of control.** Under suitable conditions on the spatial operator A where spectral decomposition is available, we are interested in the control of a increasing sequence of finite dimensional projections which approximates our problem. We start by studying finite dimensional systems.

Suppose that the linear system 1.1 is controllable. Hence for all $T > 0$ and $y_0 \in \mathbb{R}^n$, there exists at least one control u such that the corresponding solution satisfies $y(T) = 0$.

Our objective is to quantify the cost of null contrabillity. To do this we focus on the worst case: the largest size of a control needed to drive the state to zero at time T, among every initial condition y_0 on the unit ball. It is expectable for this cost to increase as $T \to 0$.

In order the transfer information from the finite-dimensional projectios to the former PDE problem, we need to understand the asymptotic behavior of this cost as $T \to 0$. This step is also important to transfer information from linear PDEs to nonlinear ones.

Denote by $|\cdot|$ the Euclidean norm on $\mathbb{R}^n$. For $T > 0$, define the null control as

Definition 1.2. *For $T > 0$, we define the null-control cost as*

$$\mathcal{K}^p(T) := \sup_{y_0\in\mathbb{R}^n,\ |y_0|=1} \inf\{\|u\|_{L^p} \mid u \in L^p(0, T; \mathbb{R}^m),\ y(0) = y_0,\ y(T) = 0\}. \tag{1.2}$$

Remark 1.1. *Note that in the previous definition we fidex the given dimension $n \in \mathbb{N}$. We will see a way to estimate the null-control cost by its inscreasingly larger finite dimensional projections.*

Proposition 1.1. *Let $0 \le K \le n-1$ be the minimal integer such that Kalman's criterion is satisfied:*

$$\operatorname{rank}[B, AB, A^2B, \dots, A^KB] = n.$$

Then, there exists $c > 0$ such that

$$\mathcal{K}^p(T) \sim cT^{-K-1+1/p},$$

as $T \to 0$.

Proof: We follow several steps.

Step 1: *Time scaling.* Given $u \in L^p(0, T; \mathbb{R}^m)$, we introduce $v(s) := u(sT)$. From Duhamel's fórmula at time $t = T$, the condition $y(T) = 0$ is equivalent to

$$F_T(v) := \int_0^1 e^{-s(TA)} B v(s) ds = -\frac{y_0}{T}. \tag{1.3}$$ {{FTv}}

Since $\|u\|_{L^p(0,T)} = T^{1/p}\|v\|_{L^p(0,1)}$, we obtain that

{ScalingT}} $$\mathcal{K}^p(T) = T^{1/p-1}\mathcal{K}(T), \quad \text{where} \quad \tilde{\mathcal{K}}(T) := \sup_{z\in\mathbb{R}^n,\ |z|=1} \inf\{\|v\|_{L^p} \mid F_T(v) = z\}. \tag{1.4}$$

Step 2: *Decomposition of the state space.* Write $B = (b_1, \dots, b_m)$. For $0 \le k \le K$, define $S_k := \operatorname{span}\{A^j b_l \mid 0 \le j \le k,\ 1 \le l \le m\} \subset \mathbb{R}^n$ and E_k the orthogonal complement of S_{k-1} in S_k with the convention $S_{-1} = \{0\}$. By assumption $E_0 \oplus \cdots \oplus E_K = \mathbb{R}^n$ and $E_K \neq \{0\}$. Let $\mathbb{P}_k$ the orthogonal projection on E_k, then $\mathbb{P}_k A^j b_l = 0$ for any $j < k$.
Finally, for $z \in \mathbb{R}^n$ define $\Delta_T(z) := \sum_{k=0}^K T^{-k}\mathbb{P}_k z$ with inverse $\Delta_T^{-1}(z) := \sum_{k=0}^K T^k \mathbb{P}_k z$.

Step 3: *Asymptotic expansion of F_T.* Expanding the exponential in 1.3 we get $\Delta_T F_T = \Lambda + R_T$. Indeed, since

$$e^{-sTA} = \sum_{j=0}^{\infty} \frac{(-sTA)^j}{j!},$$

and using that

$$Bv(s) = \sum_{l=1}^m b_l v_l, \qquad \mathbb{P}_k(A^j b_l) = 0, \quad \forall j < k.$$

Inserting this, we get $\Lambda(v)$ for $j = k$ and $R_T(v)$ for $j \ge k+1$ as follows

$$\Lambda(v) := \sum_{k=0}^K \sum_{l=1}^m \mathbb{P}_k(A^k b_l) \int_0^1 \frac{(-s)^k}{k!} v_l(s) ds$$
$$R_T(v) := \sum_{k=0}^K \sum_{l=1}^m \sum_{j=k+1}^{\infty} \frac{\mathbb{P}_k(A^j b_l)}{j!} T^{j-k} \int_0^1 (-s)^j v_l(s) ds,$$

where $v \in L^p(0,1;\mathbb{R}^m)$. In particular, F_T scales like T^k on each E_k. For $T \le 1$ we get

$$|R_T(v)| \le (K+1) \sum_{l=1}^m \sum_{j=0}^{+\infty} \frac{\|A\|^j |b_l|}{j!} T \|v_l\|_{L^1} \le T C_R \|v\|_{L^p}. \tag{1.5}$$

Step 4: $\Lambda : L^p(0,1;\mathbb{R}^m) \to \mathbb{R}^n$ *is surjective.* We construct a pregimage for Λ. Since each E_k is spanned by the $\mathbb{P}_k(A^k b_l)$, for each $z \in \mathbb{R}^n$, there exists $\alpha_0, \dots, \alpha_K \in \mathbb{R}^m$ such that $z = \sum_{k=0}^K \mathbb{P}(A^k B \alpha_k)$. Moreover, for each $0 \le k \le K$ there exists $\varphi_k \in C^\infty([0,1];\mathbb{R})$ such that $\int_0^1 (-s)^j \varphi_k(s) ds = j! \mathbf{1}_{j=k}$ for all $0 \le j \le K$. Thus using $v(s) := \sum_{k=0}^K \varphi_k(s)\alpha_k$ we get that $\Lambda(v) = z$.

In particular, there exists $U \in \mathcal{L}(\mathbb{R}^n; L^p((0,1);\mathbb{R}^m))$ such that $\Lambda \circ U = \mathrm{Id}_{\mathbb{R}^n}$ and $C_\alpha := \|U\| < \infty$.

Step 5: *Definition of the constant.* Since Λ is bounded and $E_K \neq \{0\}$,

nitionOfC}} $$c := \sup_{|z|=1, z\in E_K} \inf\{\|v\|_{L^p} \mid \Lambda(v) = z\} \in (0, C_\Lambda]. \tag{1.6}$$

For all $\varepsilon > 0$ and $z \in E_K$, there exists a local inverse $v := V_\varepsilon(z)$ such that $\|V_\varepsilon(z)\| \le (c+\varepsilon)|z|$ and $\Lambda(V_\varepsilon(z)) = z$.

Step 6: *Upper bound on the cost.* For all $\varepsilon > 0$, there exists $T_\varepsilon > 0$ such that, for all $T \in (0, T_\varepsilon]$ and all $z \in \mathbb{R}^n$, there exists $v \in L^p((0,1);\mathbb{R}^m)$ such that $F_T(v) = z$ and $\|v\|_{L^p} \le (c + 2\varepsilon) T^{-K}|z|$.

For $T, \varepsilon > 0$ and $z \in \mathbb{R}^n$, we can decompose $z = z_K + z'$ where $z_K \in E_K$ and $z' \in E_K^\perp$. To solve $F_T(v) = z$ we set $v := \sum_{i=0}^\infty v_i$ where we define

$$v_0 := V_\varepsilon(\Delta_T(z_K)) + U(\Delta_T(z')),$$

and $v_{i+1} = -U(R_T(v_i))$. Then by recursion

$$\|v_0\| \le (c+\varepsilon)T^{-K}|z_K| + C_\Lambda T^{-K+1}|z'| \le (c+\varepsilon+C_\Lambda T)T^{-K}|z|.$$

Moreover, since

$$\|v_{i+1}\| \le C_\Lambda T C_R \|v_i\|,$$

we obtain a normal convergent series and

$$\|v\| \le (c+\varepsilon+TC')T^{-K}|z|.$$

By telescoping sums, $F_T(v) = z$ and choosing T_ε small enough, we obtain $\|v\| \le (c+2\varepsilon)T^{-K}|z|$.

Step 7: *Lower bound on the cost.* For all $\varepsilon > 0$, there exists $T_\varepsilon > 0$ such that, for all $T \in (0, T_\varepsilon]$, there exists $z \in \mathbb{R}^n$ with $z \neq 0$ such that for all $v \in F_T^{-1}(z)$, one has $\|v\|_{L^p} \ge (c-2\varepsilon)T^{-K}|z|$.

By 1.6, given $\varepsilon > 0$, there exists $z^* \in E_K$ with $z^* \neq 0$ such that $\Lambda(v) = z^*$ implies $\|v\|_{L^p} \ge (c-\varepsilon)|z^*|$. For $T > 0$, set $z_T := \Delta_T^{-1}(z^*)$ and let v_T such that $F_T(v_T) = z_T$. Thus $(\Lambda + R_T)(v_T) = z^*$.

Let $w_T := U(R_T(v_T))$ so $\Lambda(w_T) = R_T(v_T)$. Then $\Lambda(v_T + w_T) = z^* \in E_K$. Hence $\|v_T + w_T\| \ge (c-\varepsilon)|z^*|$. But since $\|w_T\| \le C_\Lambda T C_R \|v_T\|$, we have that $(1 + TC_\Lambda C_R)\|v_T\| \ge (c-\varepsilon)|z^*| = (c-\varepsilon)T^{-K}|z_T|$. For $T \in (0, T_\varepsilon]$ where $T_\varepsilon := \varepsilon/(C_\Lambda C_R(c-2\varepsilon))$, we have found $z_T \in \mathbb{R}^n$ with $z_T \neq 0$ such that any v_T satisfying $F_T(v_T) = z_T$ satisfies $\|v_T\| \ge (c-2\varepsilon)T^{-K}|z_T|$.

Step 8: *Conclusion.* We proved $\tilde{\mathcal{K}} \sim cT^{-K}$. Together with 1.4 we obtain the result. □

{Prop1.9}

Proposition 1.2. *For all $T > 0$ and $n \ge 1$,*

$$\mathcal{K}_n(T) \le \sqrt{\frac{n^2}{T}} \exp\left(6\sqrt{\frac{n}{T}}\right). \tag{1.7}$$

2. From linear ODEs to linear dissipative PDEs

Now we focus on infinite dimensional control systems assuming the controllability of some projections of the state onto finite dimensional spaces and that the original system es small-time null controllable.

We will consider only the null controllability problem. Heuristically, one can think that we consider systems of the form

$$\dot{y} = Ay + Bu \tag{2.1}$$

{{HeurSystem}}

where A and B are operatores which generalizes the matrices of 1.1 and we aim for $y(T) = 0$.

2.1. Input-Output formalism.

We will present the time-iteration argument using input-output formalism so we can avoid operator semigroups theory. Let Y be a Banach space known as the state space of possible values for $y(t) \in Y$. In the same way, let U be a Banach space known as the control space describind the possible values for $u(t) \in U$. Fix $p \in [1, \infty]$.

Definition 2.1. *For $u, v \in L^p(\mathbb{R}_+, U)$, denote by $u \diamond_\tau v \in L^p(\mathbb{R}_+; U)$ their $\tau-$concatenation*

$$(u \diamond_\tau v)(s) := \begin{cases} u(s), & s \in [0, \tau), \\ v(s-\tau), & s \ge \tau. \end{cases}$$

{AbstractLinea

Definition 2.2. *We say that Σ is an abstract linear control system when it is a family $(\Sigma_t)_{t\ge 0}$ of continuous linear maps from $Y \times L^p(\mathbb{R}_+; U)$ to Y such that:*

(1) *for all $y_0 \in Y$ and $u \in L^p(\mathbb{R}_+;U)$, the map $t \mapsto \Sigma_t(y_0,u)$ is continuous on $\mathbb{R}_+$ wiht $\Sigma_0(y_0,u) = y_0$,*
(2) *for all $t,\tau \geq 0$, $y_0 \in Y$ and $u,v \in L^p(\mathbb{R}_+;U)$,*

$$\Sigma_{t+\tau}(y_0, u \diamond_\tau v) = \Sigma_t(\Sigma_\tau(y_0,u),v).$$

In other words, $\Sigma_t(y_0,u)$ represents the state $y(t) \in Y$ at time $t \geq 0$ being controlled by $u \in L^p(\mathbb{R}_+;U)$ from an initial condition y_0 governed by a PDE.

{2.3}

Lemma 2.1. *Let $T > 0$. There exists $M_T > 0$ such that $\forall t \in [0,T]$, $\|\Sigma_t\| \leq M_T$.*

Proof: It suffices to note that first point of 2.2 proves continuity, which bound uniformly $\Sigma_t(y_0,u)$. The conclusion comes from Banach-Steinhaus theorem. □

Remark 2.1. *We will only consider time-invariant systems, but the definition 2.2 can we adapted to more general families of propagators.*

2.2. **A general time iteration.** Lebeau-Robbiano strategy is a time-iteration argument wich establishes the small-time null controllability of a dissipative PDE from the balance between the cost of the small-time null controllability of its projections and its dissipation rate.

In this section we present a abstract version of their argument which applies directly.

ssumption1}

Assumption 1. *There exists a family $(P_k)_{k\in\mathbb{N}}$ of proyections on Y and there exists constantes $C_1, C_2, c_1, c_2, a, b, m > 0$ with $a < b$ such that, for all $k \in \mathbb{N}$,*

- ***Control of low modes.*** *For all $T \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^p(\mathbb{R}_+;U)$ such that*

lLowModes}}
$$P_k \Sigma_T(y_0,u) = 0 \quad \text{and} \quad \|u\|_{L^p} \leq C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{2.2}$$

- ***Dissipation of high modes.*** *For all $T \in (0,1]$ and $y_0 \in Y$,*

HighModes}}
$$P_k y_0 = 0 \quad \Rightarrow \quad \|\Sigma_T(y_0,0)\|_Y \leq C_2 e^{-c_2 T^m k^b} \|y_0\|_Y. \tag{2.3}$$

We would like to partition the time interval to apply one of the previous estimates. In each step of duration τ, we choose a treshold $k \in \mathbb{N}$, control the low modes to zero on $[0,\tau/2]$ and then let the energy dissipate on $[\tau/2,\tau]$. More precisely, the main time iteration relies on the following one-step contraction result.

{2.6}

Lemma 2.2. *Suppose that Assumption 1 holds. Then there exists $C_3, c_3 > 0$ such that, for all $k \in \mathbb{N}$, for all $\tau \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^p(\mathbb{R}_+;U)$ such that*

$$\|u\|_{L^p} \leq C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{2.4}$$

$$\|\Sigma_\tau(y_0,u)\|_Y \leq C_3 e^{c_1 k^a} e^{-c_3 \tau^m k^b} \|y_0\|_Y. \tag{2.5}$$

Proof: Assume Assumption 1 and use the same constants therein. Let $\tau > 0$, $k \in \mathbb{N}$ and $Y_0 \in Y$. Define a control u in $[0,\tau]$ as given in 2.2 for a time $\tau/2$, extended by 0 on $[\tau/2,\tau]$. Let $y_1 := \Sigma_{\tau/2}(y_0,u)$. By Proposition 2.1 and 2.2,

$$\|y_1\|_Y \leq M_1(\|y_0\|_Y + \|u\|_{L^p}) \leq M_1(1 + C_1 e^{c_1 k^a})\|y_0\|_Y.$$

Since $P_k y_1 = 0$, 2.3 implies that

$$\|\Sigma_\tau(y_0,u)\|_Y = \|\Sigma_{\tau/2}(y_1,0)\| \leq C_2 e^{-c_2(\tau-\tau_2)^m k^b} \|y_1\|_Y,$$

we deduce the claimed estimates with $c_3 = c_2 2^{-m}$ and $C_3 = M_1 C_2(1 + C_1)$. □

Theorem2.7}

Theorem 2.1. *Suppose that Assumption 1 holds. Then the abstract linear system Σ is small-time null-controllable. More precisely, there exists constantes $C_0, c_0 > 0$ such that, for every $T \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^p((0,T);U)$ such that $\Sigma_T(y_0,u) = 0$ and*

lEstimate}}
$$\|u\|_{L^p} \leq C_0 \exp\left(c_0 T^{-\sigma}\right) \|y_0\|_Y, \qquad \sigma := \frac{am}{b-a}. \tag{2.6}$$

Proof: We separate the proof in several steps.

Step 1. *Construction of the time and frequency sequences.* We define a sequence of time intervals and frequency cutoffs. Let $r > 1$ and $q > 1$ be two real numbers to be chosen later. We decompose the interval $[0, T]$ into series of subintervals. Let

$$\alpha := T\left(1 - \frac{1}{r}\right), \quad \text{and} \quad \tau_j := \frac{\alpha}{r^j}, \quad j \in \mathbb{N}. \tag{2.7}$$ {{2.11}}

Hence $T = \sum_{j=0}^{\infty} \tau_j$. We define a sequence of time instants $T_0 = 0$ and $T_{j+1} = T_j + \tau_j$. The j-step of our procedure will happen on the interval $[T_j, T_{j+1}]$.

Let $\beta > 0$ be another parameter to be chosen. We define the sequence of frequency cutoffs as

$$k_j := \lfloor \beta q^j \rfloor, \quad \forall j \in \mathbb{N}. \tag{2.8}$$

Step 2. *The iteration scheme.* We define a sequence of states $(y_j)_{j\in\mathbb{N}}$ by y_0 the given initial estimate and $y_{j+1} := \Sigma_{\tau_j}(y_j, u_j)$ where u_j is the control used on $[T_j, T_{j+1}]$ given by Proposition 2.2. Hence, by the estimates from 2.2 we get

$$\|u_j\|_{L^p} \leq C_1 e^{c_1 k_j^a} \|y_j\|_Y, \tag{2.9}$$ {{IterativeSch
$$\|y_{j+1}\|_Y \leq C_3 e^{-\gamma_j} \|y_j\|_Y, \tag{2.10}$$

where we introduced the notation $\gamma_j := c_3 \tau_j^m k_j^b - c_1 k_j^a$.

Step 3. *Choice of the parameters r, q, β.* Our goal is to ensure that $\|y_j\|_Y \to 0$ sufficiently fast. For $\beta \geq 1$, by the definition of k_j we know that $\beta q^j/2 \leq k_j \leq \beta q^j$. Recalling 2.7 for τ_j, we obtain

$$\gamma_j \geq (q^a)^j \left(2^{-b} c_3 \alpha^m \beta^b (q^{b-a} r^{-m})^j - c_1 \beta^a\right). \tag{2.11}$$

We let $q := 2^{1/a}$ so that $q^a = 2$ and $r := q^{\frac{b-a}{m}}$, hence $q^{b-a} r^{-n} = 1$. Also, we tune β such that $2^{-b} c_3 \alpha^m \beta^b = 3 c_a \beta^a$, this corresponds to setting

$$\beta := \left(\frac{2^b 3 c_1}{c_3 \alpha^m}\right)^{\frac{1}{b-a}}. \tag{2.12}$$ {{2.17}}

With these choices $\beta \to +\infty$ as $T \to 0$ and one has

$$\gamma_j \geq 2 c_1 \beta^a 2^j.$$

Step 4. *Stepwise convergence of the state.* Iterating the construction from step 2, we obtain for any $j \geq 1$,

$$\textbf{??} \|y_j\|_Y \leq \exp\left(j \ln C_3 - 2 c_1 \beta^a (2^j - 1)\right) \|y_0\|_Y. \tag{2.13}$$

Therefore $\|y_j\|_Y \to 0$ as $j \to \infty$.

Step 5. *Bounding the total control cost.* The overall control u is teh concatenation of the controls u_j on $[T_j, T_{j+1}]$. Using 2.9 and **??**,

$$\begin{aligned}
\|u_j\|_{L^p} &\leq C_1 e^{c_1 k_j^a} \|y_j\|_Y \\
&\leq C_1 \exp(c_1 \beta^a 2^j) \exp(j \ln C_3 - 2 c_1 \beta^a (2^j - 1)) \|y_0\|_Y \\
&\leq C_1 e^{c_1 \beta^a} \exp(j(\ln C_3 - c_1 \beta^a)) \|y_0\|_Y \\
&\leq C_1 e^{c_1 \beta^a} 2^{-j} \|y_0\|_Y.
\end{aligned}$$

Using $2^j - 1 \geq j$ and T small enough so taht $\ln C_3 - c_1\beta^a \leq -\ln 2$. Thus, there exists $T_0 > 0$ such that for all $0 < T \leq T_0$,

$$\|u\|_{L^p((0,T);U)} \leq \sum_{j=0}^{\infty} \|u_j\|_{L^p}((0,\tau_j);U) \leq 2C_1 e^{c_1\beta^a}\|y_0\|_Y. \tag{2.14}$$

Recalling 2.7 and 2.12, this entails the cost estimate 2.6 since $\beta^a \approx (\alpha^{-\frac{m}{b-a}})^a \approx T^{-\sigma}$.
Step 6. *Conclusion.* Since Σ is an abstract linear control system, the associated solution $y(t) := \Sigma_t(y_0, u)$ belongs to $C^0([0,T];Y)$. By the composition property of the $\tau-$concatenation,$y(T_j) = y_j$. We proved that $y_j \to 0$. Since $T_j \to T$, by continuity we conclude that $\Sigma_T(y_0, u) = y(T) = 0$. □

We give some remakrs

Remark 2.2 (Cost optimization)**.** *In some contexts is important to optimize the constant c_0 inside the exponential of the cost estimate 2.6.*

Remark 2.3 (Time-dependent systems)**.** *The proof given above applies for time-dependent systems.*

elaxedCost}

Remark 2.4 (Relaxed assumptions on constants)**.** *In 2.2 and 2.3 we use an exponential constant to fix ideas, but is usual to find prefactors of the form T^{-h}.*

When $T^m k^b \sim k^a$, the previous result still holds when we replace 2.2 and 2.3 by

$$\|u\|_{L^p} \leq C_1 \exp(c_1 k^a + c_1' T^{-\theta})\|y_0\|_Y, \tag{2.15}$$

$$\|\Sigma_T(y_0, 0)\|_Y \leq C_2 \exp(-c_2 T^m k^b + c_2' k^a + c_2'' T^{-\theta})\|y_0\|. \tag{2.16}$$

Then the cost estimate 2.6 hold with $\sigma = \max\{\frac{am}{b-a}, \theta\}$.

Remark 2.5. *The estimate 2.2 states that the cost of control of the first k modes does not blow up as $T \to 0$.*

2.3. **The role of spectral inequalities.** The time-iteration argument from previous section is sufficiently abstract to be applied as a black box thanks to Assumption 1. In this section we explain why many authors rely on spectral inequalities to check the properties 2.2 and 2.3. There are even some cases where it is not required that the projections P_k are orthogonal projections on spectral subspaces of the considered self-adjoint operator.

During this section we will suppose that U and Y are Hilbert spaces. Consider a family $(P_k)_{k\in\mathbb{N}}$ of orthogonal projections on Y. For $t \geq 0$ define $\mathbb{T}_t \in \mathcal{L}(Y)$ by $\mathbb{T}_t y_0 := \Sigma_t(y_0, 0)$. Finally, assume that there exists an operator $B \in \mathcal{L}(U, Y)$ such that, for all $u \in L^p(\mathbb{R}_+; U)$ and $T \geq 0$,

{{2.25}}

$$\Sigma_T(0, u) = \int_0^T \mathbb{T}_{T-t} B u(t) dt. \tag{2.17}$$

When $p < +\infty$, the existence of an integral representation of $u \mapsto \Sigma_T(0, u)$ is always true, but B is not always bounded from U to Y (this usually fails for boundary controls).

The constructions of this section are based on the following static assumption, in which neither the time nor the operator A play a role.

Inequality}

Assumption 2 (Spectral inequality)**.** *There exists $C_1, c_1, a > 0$ such that, for all $k \in \mathbb{N}$,*

$$\forall y \in Y, \quad \|P_k y\|_Y \leq C_1 e^{c_1 k^a}\|B^* P_k y\|_U, \tag{2.18}$$

or, equivalently, by the closed-graph theorem,

$$\exists D_k \in \mathcal{L}(Y, U), \text{ such that } P_k = P_k B D_k \text{ and } \|D_k\| \leq C_1 e^{c_1 k^a}. \tag{2.19}$$

2.3.1. ***The commutative case.*** In this section we study the case where the projections P_k and the dynamics $\mathbb{T}_t$ commute. In particular we show that the spectral inequality entails the controllability of the low modes.

{Prop2.13}

Proposition 2.1. *Under Assumption 2, if the projections P_k commute with the dynamics, then, for all $k \in \mathbb{N}$, $T \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^p((0,T);U)$ such that*

$$P_k \Sigma_T(y_0, u) = 0, \quad \text{and} \quad \|u\|_{L^p} \leq T^{1/p-1} M_1 C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{2.20}$$

where M_1 is the constant from Lemma 2.1.

Proof: Suppose that 2 is valid. Let $k \in \mathbb{N}$ and $D_n \in \mathcal{L}(Y,U)$ be given by the Spectral Inequality. For $T \in (0,1]$ and $y_0 \in Y$, set

$$u(t) := -\frac{1}{T} D(k) T_t y_0. \tag{2.21}$$

By Lemma 2.1, we have the estimate

$$\|u\|_{L^p} \leq T^{1/p-1} \|D_k\| \cdot \sup_{t\in[0,1]} \|\mathbb{T}_t\| \cdot \|y_0\|_Y \leq T^{1/p-1} M_1 C_1 e^{c_1 k^a} \|y_0\|_Y. \tag{2.22}$$

By linearity and 2.17,

$$\Sigma_T(y_0, u) = \mathbb{T}_T y_0 - \frac{1}{T} \int_0^T \mathbb{T}_{T-t} B D_k \mathbb{T}_t y_0 dt. \tag{2.23}$$

{{2.31}}

By assumption, we know that $P_k \mathbb{T}_\tau P_k = P_k \mathbb{T}_\tau$ for every $\tau \geq 0$. Applying P_k to 2.23 and using $P_k B D_k = P_k$, the semigroup theory and the previous commutative property implies

$$\begin{aligned} P_k \Sigma_T(y_0, u) &= P_k \mathbb{T}_T y_0 - \frac{1}{T} P_k \int_0^T \mathbb{T}_{T-t} P_k B D_k \mathbb{T}_t y_0 dt \\ &= P_k \mathbb{T}_T y_0 - \frac{1}{T} P_k \int_0^T \mathbb{T}_{T-t} P_k \mathbb{T}_t y_0 dt \\ &= P_k \mathbb{T}_T y_0 - \frac{1}{T} P_k \int_0^T \mathbb{T}_{T-t} \mathbb{T}_t y_0 dt = 0. \end{aligned}$$

□

So we have found a control of the expected size, bringing the projection P_k of the state to 0.

Remark 2.6. *The above proof only uses commutation and projection property $P_k^2 = P_k$, which is equivalent to the condition that the range of $P_k^\perp = \mathrm{Id} - P_k$ is stable by the semigroup. This is slightly weaker than requiren that P_k commutes with the semigroup. One must only prevent high-to-low energy transfers.*

2.3.2. ***The non-commutative case.*** When projections P_k do not commute with $\mathbb{T}_t$, the previous construction no longer yields a null-control for the low modes. However, we can compesate the lack of commutation using dissipation to derive the one-step contraction from Lemma 2.2, allowing us to execute the time iteration of ...

{2.15}

Proposition 2.2. *Assume that there exist $C_1, C_2, c_1, c_2, a, b, m > 0$ with $a < b$ such that, for all $k \in \mathbb{N}$, the spectral inequality from Assumption 2 and the dissipation condition 2.3 hold. Then there exists $C_3, c_3 > 0$ such that, for all $k \in \mathbb{N}$, for all $k \in \mathbb{N}$, for all $\tau \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^p(\mathbb{R}_+;U)$ with*

$$\|u\|_{L^p} \leq 2\tau^{1/p-1} M_1 C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{2.24}$$

$$\|\Sigma_\tau(y_0, u)\|_Y \leq C_3 e^{c_1 k^a} e^{-c_3 \tau^m k^b} \|y_0\|_Y, \tag{2.25}$$

where M_1 is the constant from Lemma 2.1.

Proof: Let $k \in \mathbb{N}$ and $D_k \in \mathcal{L}(Y,U)$ given by Proposition 2 on $(0, \tau/2)$, which is 0 on $(\tau/2, \tau)$:

$$u(t) := -\frac{2}{\tau} \mathbf{1}_{(0,\tau/2)}(t) D_k \mathbb{T}_t y_0. \tag{2.26}$$

By Lemma 2.1, we obtain the first estimate. By linearity and 2.17,

$$\Sigma_\tau(y_0,u) = \mathbb{T}_t y_0 - \frac{2}{\tau}\int_0^{\tau/2} \mathbb{T}_{\tau-T} B D_k \mathbb{T}_t y_0 dt. \tag{2.27}$$

Then we write the following decomposition

$$BD_k = \mathrm{Id} + (BD_k - \mathrm{Id}) = \mathrm{Id} + P_k(BD_k - \mathrm{Id}) + (\mathrm{Id} - P_k)(BD_k - \mathrm{Id}). \tag{2.28}$$

By construction, $P_k(BD_k - \mathrm{Id}) = 0$. By linearity and the semigroup property,

$$\Sigma_\tau(y_0,u) = -\frac{2}{\tau}\int_0^{\tau/2} \mathbb{T}_{\tau-t}(\mathrm{Id} - P_k)(BD_k - \mathrm{Id})\mathbb{T}_t y_0 dt. \tag{2.29}$$

Hence, by the dissipation condition 2.3,

$$\begin{aligned}\|\Sigma_\tau(y_0,u)\|_Y &\le \frac{2}{\tau}\int_0^{\tau/2} \|\mathbb{T}_{\tau-t}(\mathrm{Id} - P_k)\|(\|B\|\|D_k\| + 1)\|\mathbb{T}_t\|\|y_0\|_Y dt \\ &\le C_2 e^{-c_2(\tau/2)^m k^b}(\|B\|C_1 e^{c_1 k^a} + 1)M_1\|y_0\|_Y.\end{aligned}$$

This we obtain the last estimate with $C_3 = C_2(C_1\|B\| + 1)M_1$ and $c_3 = c_2 2^{-m}$. □

2.4. **Historical perspective.** Here we recall some contributions regarding the method exposed above, commonly referred to as the "Lebeau-Robbiano method".

The 1995 paper of Lebeau and Robbiano. To prove the small-time null controllability of the heat equation posed on a smooth compact Riemannian manifold $\mathcal{M}$, the strategy is as follows.

(1) Consider the family of eigenfunctions $(e_k)_{k\in\mathbb{N}}$ of the Dirichlet Laplacian $A = -\Delta$, with associated eigenvalues $(\lambda_k)_{k\in\mathbb{N}}$. Consider P_k to be the projection on the finite-dimensional space spanned by $e_1,\dots,e_k$.
(2) The previous automatically ensures that $\|\Sigma_T(y_0,0)\|_Y \le e^{-\lambda_k T}\|y_0\|_Y$ when $P_k y_0 = 0$. By Weyl's law, this entails the dissipation property 2.3 with $b = 2/d$ and $m = 1$.
(3) Using Carleman estimates for the elliptic operator $\partial_t^2 + \Delta$, prove that for any $\theta > 1$, for any $T \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^\infty(0,T)$ such that $P_k\Sigma_T(y_0,u) = 0$ with a cost estimate of the form

$$\|u\|_{L^\infty} \le C_1 \exp\left(c_1\lambda_k^{1/2} + c_1 T^{-\theta}\right)\|y_0\|_Y. \tag{2.30}$$

This corresponds to our generalized cost assumption with $a = 1/d$ by Weyl's law.

More recent variations and enhancements. The Lebeau-Robbiano strategy has been applied in countless settings and many authors have proposed improvements.

- In ... Jerison, Lebeau and Zuazua started to isolate and study precisely static spectral inequalities of the form of Assumption 2, avoiding the lossy estimate about the control 2.6.
- In ... Miller started the first abstraction of the previous results which concerned the heat equation to self-adjoint operators A with admissible control operators B, obtaining the qualitatively optimal cost as $T \to 0$ for the control of a family of diffusion equations.
- Subsequent works improved the estimates of the cost of controllability or reduced the assumptions on the control operator B. Miller also remarked that commutation is not necessary: we only need stability of the high modes as in Remark 2.14
- Beauchard, Pravda-Starov and Miller introduced the key novelty that projections P_k don't need to be spectral projections onto the eigenspaces of the operator A, which means that it is not required to be a self-adjoint operator. This important improve allows to reuse well-known spectral inequalities for a reference operator and apply it to prove the controllability of another PDE.
- Later it was identifies that the method can be extended to time-dependent systems (with time-dependent operators or time-dependent control domains).

3. Example: interior contrl of the heat equation on $[0,1]$.

3.1. **Comments on the formalism.**

3.1.1. *Time continuity and comparison with the usual definition.*

3.2. **Why only null-controllability?** First, for ODEs and some conservative PDEs such as the semilinear wave equation, one can often prove exact controllability.

The problem for dissipative PDEs studied in this section is the regularizing effect of the dynamics. This means that we cannot hope for an exact controllability result since as soon as $t > 0$, solutions become smoother that the initial data even with a non-zero control. At a set level, for fixed $T > 0$ the space of reachable states is a proper subspace of the state space Y.

Example 3.1.

Null controllability and controllability to trajectories. In more complex PDE models, a common and robust objective is controllability to trajectories. Fix a reference pair $(\overline{y_0}, \overline{u})$ and ley $\overline{y}$ be te associated solution. One asks wheter, for any other initial state $y_0 \in Y$, there exists a control $u \in L^p((0,T);U)$ such that $y(T) = \overline{y}(T)$. The special case $\overline{y_0} = 0$, $\overline{u} = 0$ is the null controllability problem.

For linear systems, controllability to trajectories is equivalent to null controllability.

Operator theoretic definition.
We can formalize these notion using input-output maps. For $T > 0$, recall that the semigroup $\mathbb{T}_t : Y \to T$ and the control map $\Phi_T : L^p((0,T);U) \to Y$ are defined by $\mathbb{T}_T y_0 := \Sigma_T(y_0, 0)$ and $\Phi_T y := \Sigma_T(0, u)$. By linearity, the final state is given by $y(T) = \mathbb{T}_T y_0 + \Phi_T u$, thus

- *Exact controllability.* Is the sourjectivity condition $\operatorname{Ran} \Phi_T = Y$.
- *Null controllability.* Is the inclusion $\operatorname{Ran} \mathbb{T}_T \subset \operatorname{Ran} \Phi_T$.
- *Approximate controllability.* Is the density condition $\overline{\operatorname{Ran} \Phi_T} = Y$.

For dissipative systems, since $\mathbb{T}_T$ maps Y into a smaller subspace of smooth functions, the inclusion $\operatorname{Ran}\mathbb{T}_\mathrm{t} \subset \operatorname{Ran} \Phi_\mathrm{T}$ es much easier to satisfy than $\operatorname{Ran} \Phi_T = Y$.

3.3. **Example: interior control of the heat equation on [0,1].** We investigate the small-time null controllability of the heat equation on the segment [0,1], with an interior control. More precisely, for $T > 0$, we consider the system

$$\begin{cases} y_t - y_{xx} = u 1_\omega, \quad (0,T) \times (0,1), \\ y(t,0) = y(t,1) = 0, \quad (0,T), \\ y(0,x) = y_0(x), \quad (0,1). \end{cases} \tag{3.1}$$ {{InnerHeatEq}}

where $y_0 \in L^2(0,1)$ is the initial data and $u \in L^2((0,T) \times \omega)$ is a control, supported in a fixed measurable subset $\omega \subset (0,1)$. We will consider solutions to this system in the space

$$\mathcal{Y} := C^0([0,T]; L^2(0,1)) \cap L^2((0,T); H_0^1(0,1)) \cap H^1((0,T); H^{-1}(0,1)),$$

equipped with the canonical norm. The general well-posedness result is

Proposition 3.1. *For any $T > 0$, $y_0 \in L^2(0,1)$ and $f \in L^2((0,T); H^{-1}(0,1))$, there exists an unique solution $y \in \mathcal{Y}$ to*

$$\begin{cases} y_t - y_{xx} = f, \quad (0,T) \times (0,1), \\ y(t,0) = y(t,1) = 0, \quad (0,T), \\ y(0,x) = y_0(x), \quad (0,1). \end{cases} \tag{3.2}$$

Moreover, there exists a constant $C > 0$ such that for every T, f, y_0,

$$\|y\|_{C^0 L^2} + \|y\|_{L^2 H^1} + \|y\|_{H^1 H^{-1}} \leq C(\|y_0\|_{L^2} + \|f\|_{L^2 H^{-1}}).$$

3.3.1. *Energy dissipation.* Let $(\varphi_j = \sqrt{2}\sin(j\pi x))_{j\in\mathbb{N}}$ be an orthonormal basis of $L^2(0,1)$ of eigenfunctions of $-\partial_{xx}$ with homogeneous Dirichlet conditions. The eigen values came of the form $\lambda_j = (j\pi)^2$. Decomposing $y(t)$ on this basis, the explicit resolution with $u = 0$ leads to

$$\langle y(t), \varphi_j\rangle = e^{-(j\pi)^2 t}\langle y_0, \varphi)j\rangle. \tag{3.3}$$

In particular, if $\langle y_0, \varphi_j\rangle = 0$ for $1 \le j \le k$, one has

$$\|y(t)\|_{L^2} \le e^{-(k\pi)^2 t}\|y_0\|_{L^2}. \tag{3.4}$$

3.3.2. *Spectral inequality.* We assume the next spectral inequality due to Turán,

{Prop2.17}

Proposition 3.2. *Let $\omega \subset [0,1]$ be a measurable set with $|\omega| > 0$. there exists $C_\omega > 0$ such that for any $n \ge 1$ and $a_1, \dots, a_n \in \mathbb{R}$,*

$$\|f\|_{L^2(0,1)} \le C_\omega^n \|f\|_{L^2(\omega)}, \quad f(x) := \sum_{j=1}^{n} a_j \sin(j\pi x).$$

3.3.3. *Small-time null controllability.*

Theorem 3.1. *Assume that $|\omega| > 0$. System 3.1 is small-time null controllable and there exists $C > 0$ such that, for all $T > 0$ and $y_0 \in L^2(0,1)$, there exists $u \in L^2((0,T)\times\omega)$ such that the associated solution to 3.1 satisfies $y(T) = 0$ and $\|u\|_{L^2} \le ce^{C/T}\|y_0\|_{L^2}$.*

Proof: Denote by P_k the projection on $\operatorname{span}\{\varphi_1, \dots, \varphi_k\}$ in $L^2(0,1)$. By energy dissipation, 2.3 holds with $b = 2$ and $m = 1$. Since P_k commutes with $\mathbb{T}_t$ and $B \in \mathcal{L}(L^2(\omega); L^2(0,1))$ is the extension by 0, so B^* is the restriction to ω. Hence by Proposition 3.2, for any $y \in L^2(0,1)$,

$$\|P_k y\|_Y \le e^{c_\omega k}\|B^* P_k y\|_U.$$

Thus we can apply Proposition 2.1 which implies that the control cost estimate holds with $a = 1$. Thus Proposition 1 holds and Theorem 2.1 applies.

Since $p = 2$, Proposition 2.1 leads to a singular $T^{-1/2}$ prefactor, which is an absorbable weak degeneracy with the same iteration. □

3.4. **Example: Boundary control of the heat equation on $[0,\pi]$.** Here we investigate for $T > 0$ the problem

aryHeatEq}

$$\begin{cases} y_t - y_{xx} = 0, & (0,T)\times(0,\pi), \\ y(t,0) = u(t), & (0,T), \\ y(t,\pi) = 0, & (0,T), \\ y(0,x) = y_0(x), & (0,\pi). \end{cases} \tag{3.5}$$

where $y_0 \in H^{-1}(0,\pi)$ is the initial data and $u \in L^2((0,T);\mathbb{R})$ is a control. We have the next well-posedness result.

Proposition 3.3. *For any $T > 0$, $y_0 \in H^{-1}(0,\pi)$ and $u \in L^2((0,T);\mathbb{R})$, there exists a unique weak solution $y \in C^0([0,T]; H^{-1}(0,\pi)) \cap L^2((0,T); L^2(0,\pi))$ defined by transposition to 3.5.*

Once again, we denote by $(\varphi_j = \sin(jx))_{j\in\mathbb{N}}$ the orthogonal basis of $L^2(0,\pi)$ of eigenfunctions of $-\partial_{xx}$ with Dirichlet boundary conditions associated with the eigenvalues $\lambda_j = j^2$. Testing equation 3.5 against φ_j we get

$$\frac{d}{dt}\langle y, \varphi_j\rangle = ju(t) - j^2\langle y, \varphi_j\rangle,$$

so by Duhamel representation formula,

$$\langle y(T), \varphi_j\rangle = e^{-j^2 T}\langle y_0, \varphi_j\rangle + j\sqrt{2/\pi}\int_0^T e^{-j^2(T-t)}u(t)dt. \tag{3.6}$$

3.4.1. *Energy dissipation.* If $\langle y_0, \varphi_j \rangle = 0$ for all $1 \leq j \leq k$, one has

$$\|y(t)\|_{L^2} \leq e^{-k^2 t} \|y_0\|_{L^2}$$

.

3.4.2. *Cost of control of the low frequencies.* To control low frequencies we rely on the Proposition 1.2, sinche there exists an $u \in L^2((0,T);\mathbb{R})$ such that $\langle y(T), \varphi_j \rangle = 0$ for all $1 \leq j \leq k$. It means thta we have

$$\|u\|_{L^2} \leq \sqrt{\frac{k^2}{T}} \exp\left(\sqrt{\frac{k}{T}}\right) \|y_0\|_{H^{-1}}.$$

Using $x \leq e^x$ and Young inequality we obtain for $T \in (0,1]$,

$$\|u\|_{L^2} \leq e^{4(k-T^{-1})} \|y_0\|_{H^{-1}}.$$

3.4.3. *Small-time null controllability.* Typically the Lebeau-Robbiano strategy does not apply for boundary controllability since the spectral inequality required is not satisfied. However, we can use the time iteration of the Lebeau-Robbiano method and another technique to achieve the controllability of the low frequencies.

Theorem 3.2. *System 3.5 is small-time null controllable.*

Proof: Let P_k be the projections on $\operatorname{span}\{\varphi_1, \ldots, \varphi_k\}$. By energy dissipation, 2.3 holds with $b = 2$ and $m = 1$. By the previous estimate on u, the relaxed cost estimate 2.4 holds with $a = 1$ and $\sigma = 1$. Thus Assumption 1 holds and Theorem 2.1 applies. □

3.5. **Example: Control of the heat equation in $\mathbb{R}^d$ from a thick set.** In this section we investigate the problem

$$\begin{cases} y_t - \Delta y = u 1_\omega, & (0,T) \times \mathbb{R}^d, \\ y(0,x) = y_0(x), & \mathbb{R}^d. \end{cases} \tag{3.7}$$ {{RdHeatEq}}

Where $y_0 \in L^2(\mathbb{R}^d)$ is the initial data and $u \in L^1((0,T); L^2(\omega))$ a contorl supported in a fixed measurable set $\omega \subset \mathbb{R}^d$.

Lemma 3.1. *For any $T > 0$, $y_0 \in L^2(\mathbb{R}^d)$ and $u \in L^1((0,T); L^2(\omega))$, there exists an unique solution $y \in C^0([0,T]; L^2(\mathbb{R}^d))$ to system 3.7.*

3.5.1. *Energy dissipation and projections.* In absence of control, the Fourier transform reads

$$\hat{y}(t,\xi) = e^{-t|\xi|^2} \hat{y_0}(\xi).$$

In particular, if $\operatorname{supp} \hat{y_0} \subset B_{\mathbb{R}^d}(0,k)$ for some $k \geq 0$, one has

$$\|y(t)\|_{L^2} \leq e^{-tk^2} \|y_0\|_{L^2}.$$

3.5.2. *Spectral inequality.* We need to find an spectral inequality of the form

$$\|P_k y\|_{L^2(\mathbb{R}^d)} \leq C_k \|P_k y\|_{L^2(\omega)}.$$

For the system 3.7 to be small-time controlalble, it is natural to require that the control domain ω is well spread-out all over $\mathbb{R}^d$. To this, we use the next definition

Definition 3.1. *Let $\gamma \in (0,1]$ and $a > 0$. We say that a measurable $\omega \subset \mathbb{R}^d$ is a $(\gamma, a)-$thick when*

$$\forall x \in \mathbb{R}^d, \qquad \left|\omega \cap (x + [0,a]^d)\right| \geq \gamma a^d.$$

In this kind of sets we can apply the next equality.

{Prop2.23}

Proposition 3.4. *There exists an universal constant $C > 0$ such that, for any $\gamma \in (0,1]$, $a > 0$ and any measurable subset $\omega \subset \mathbb{R}^d$ which is $(\gamma, a)-$thick, one has*

$$\|f\|_{L^2(\mathbb{R}^d)} \leq \left(\frac{C}{\gamma}\right)^{C(N_a+1)} \|f\|_{L^2(\omega)},$$

for all $N \geq 0$ and $f \in L^2(\mathbb{R}^d;\mathbb{C})$ such that $\operatorname{supp}(\hat{f}) \subset B_{\mathbb{R}^d}(0,N)$.

3.5.3. *Small-time null controllability.*

Theorem 3.3. *Let $\gamma \in (0,1]$ and $a > 0$. Assume that ω is a (γ, a)-thick. Then the heat equation 3.7 is small-time null controllable.*

Proof: Define P_k as the projection of the Fourier modes $|\xi| \leq k$. More precisely, set

$$P_k y := \mathcal{F}^{-1}(\xi \mapsto \hat{y}(\xi) 1_{\{|\xi| \leq k\}}).$$

First, by energy dissipation, 2.3 holds with $b = 2$ and $m = 1$.

Second, since ω is a (γ, a)-thick set, by the proposition we have for any $y \in L^2(\mathbb{R}^d)$,

$$\|P_k y\|_{L^2(\mathbb{R}^d)} \leq C_1 e^{c_1 k} \|P_k y\|_{L^2(\omega)},$$

for some constante $C_1, c_1 > 0$ depending on a, γ. We have that P_k commutes with the semigroup thanks to the Fourier solution. Then the operator $B : L^2(\omega) \to L^2(\mathbb{R}^d)$ is the extension by 0 operator. and B^* is merely the restriction to ω and $\|P_k y\|_{L^2(\omega)} = \|B^* P_k y\|_{L^2(\omega)}$. Thus by Proposition 2.1, for anu $T \in (0,1]$ and $y_0 \in L^2(\mathbb{R}^d)$, ther eexists $u \in L^1((0,T); L^2(\omega))$ such that $P_k \Sigma_T(y_0, u) = 0$ and $\|u\|_{L^1} \leq M_1 C_1 e^{c_1 k} \|y_0\|_{L^2(\mathbb{R}^d)}$. Hence 2.2 holds with $a = 1$.

Thus Assumption 1 holds and the result follows from Theorem 2.1. □

3.6. **Example: 2D Kolmorgorov equation.** Now we investigate the small-time null controllability of a 2D Kolmogorov equation (also called Focker-Planck equation), based on the hypoelliptic operator $z\partial_{xx} - \partial_{zz}$. PDEs on this operator are ubiquitoys in kinetic theory or boundary layer theory. Consider

olmogorov}}

$$\begin{cases} y_t + z y_x - y_{zz} = u 1_\omega, & (0,T) \times \mathbb{R}^2, \\ y(0,x,z) = y_0(x,z), & \mathbb{R}^2. \end{cases} \tag{3.8}$$

where $y_0 \in L^2(\mathbb{R}^2)$ is the initial data and $u \in L^1((0,T); L^2(\omega))$ is a control suspported on a measurable subset $\omega \subset \mathbb{R}^2$.

Proposition 3.5. *For any $T > 0$, $y_0 \in L^2(\mathbb{R}^2)$ and $u \in L^1((0,T); L^2(\omega))$, there exist a unique solution $y \in C^0([0,T]; L^2(\mathbb{R}^2))$ to 3.8.*

In this case, we will use projections which are not on eigenspaces of the considered operator, which are the same projections on Fourier modes as for heat equation on the past section, re using the spectral theory for that problem involving thick sets and Logvinenko-Sereda inequality.

3.6.1. *Energy dissipation.*

Proposition 3.6. *There exists $c_0 := (4 - \sqrt{13}/6) > 0$ such that, for any $k \geq 1$ and $y_0 \in L^2(\mathbb{R}^2)$ such that $\hat{y_0} = 0$ on $B(0,k)$, we have for any $t \geq 0$,*

$$\|y(t)\|_{L^2} \leq e^{-c_0 \min\{t, t^3\} k^2} \|y_0\|_{L^2}.$$

Proof: We perform a change on variables to follow the characteristics of the Kolmogorov equation. Set $\phi(t,x,z) := y(t, x_z t, z)$. In particular, $\phi(0) = y_0$. Then since

$$\partial_t y = \partial_t \phi - z \partial_x \phi, \quad \partial_x y = \partial_x \phi, \qquad \partial_{zz} y = \partial_{zz} \phi - 2t \partial_{xz} \phi + t^2 \partial_{xx} \phi,$$

we get

$$(\partial_t - \partial_{zz} + 2t\partial_{xz} - t^2\partial_{xx})\phi = 0.$$

Hence we have a PDE with time-varying coefficients which are constant in space. Taking Fourier transform in both directions $(x, z) \mapsto (\xi, \zeta)$ we get

$$(\partial_t + \zeta^2 + 2t\xi\zeta + t^2\xi^2)\hat{\phi} = 0.$$

Explicit integration leads to

$$\hat{\phi}(t, \xi, \zeta) = \exp\left(-t\zeta^2 - t^2\xi\zeta - \frac{t^3}{3}\xi^2\right)\hat{\phi}_0(\xi, \zeta).$$

Writing

$$|t^2\xi\zeta| \leq \frac{\alpha}{2}t\zeta^2 + \frac{1}{2\alpha}t^3\xi^2,$$

and optimizing with respect to $\alpha := (2 + \sqrt{13})/3$ we obtain that

$$t\zeta^2 + t^2\xi\zeta + \frac{t^3}{3}\xi^2 \geq \frac{4 - \sqrt{13}}{6}(t\zeta^2 + t^3\xi^2).$$

Since the initial data is supported for $\zeta^2 + \xi^2 \geq k^2$ we obtain that

$$\|\phi(t)\|_{L^2} \leq e^{-c_0 \min\{1, t^2\}k^2}\|\phi(0)\|_{L^2},$$

which is the claimed estimate since for any $t \geq 0$, $\|\phi(t)\|_{L^2} = \|y(t)\|_{L^2}$.

□

3.6.2. *Small-time null controllability.*

Theorem 3.4. *Let $\gamma \in (0, 1]$ and $a > 0$. Asume that ω is (γ, a)-thick. Then the Kolmogorov equation 3.8 is small-time null controllable.*

Proof: Define P_k as the projection on the Fourier modes $|\xi|^2 + |\zeta|^2 \leq k^2$. First, by the previous estimate, the dissipation condition 2.3 holds with $m = 3$ and $b = 2$.

Second, since ω is a (γ, a)-thick, by Proposition 3.4, fir any $y \in L^2(\mathbb{R}^2)$,

$$\|P_k y\|_{L^2(\mathbb{R}^2)} \leq C_1 e^{c_1 k}\|P_k y\|_{L^2(\omega)},$$

for some constants $C_1, c_1 > 0$ depending on a, γ. Hence $B : L^2(\omega) \to L^2(\mathbb{R}^2)$ is the extension by 0 and B^* the restriction to ω with $\|P_k y\|_{L^2(\omega)} = \|B^* P_k y\|_{L^2(\omega)}$. Note that P_k does not commute with $\mathbb{T}_t$, but we can apply Proposition 2.2, then the One-step contraction applies and Theorem 2.1 ends the proof. □

4. A direct Lebeau-Robbiano strategy for the observability of heat-like semigroups

4.1. **Introduction.** This paper concerns the so-called "Lebeau-Robbiano strategy" for the null-controllability of lienar evolutions systems like the heat equation.

Here, the strategy goes directly from the observability on groeth spaces to the estimate of fast final-observability cost, and reaches the optimal exponents β for the observation from conex of heat diffusion in potential wells $V(x) = |x|^{2k}$, $k \in \mathbb{N}^*$. This can be seen has a shortcut of the original Lebeau-Robbiano strategy since it leave null-controllability out.

4.2. **Setting and main results.**

4.2.1. *Observability cost.* We consider the abstract differential equation

$$\dot{\phi}(t) = A\phi(t), \quad \phi(0) = x \in \mathcal{E}, \quad t \geq 0, \tag{4.1}$$

where $A : D(A) \subset \mathcal{E} \to \mathcal{E}$ is the generator of a strongly continuous semigroup $(e^{tA})_{t\geq 0}$ on a Hilbert space $\mathcal{E}$. The solution is $\phi(t) = e^{tA}x$. We may think of A as a non-positive self-adjoint operator with an orthonormal basis of eigenfunctions for example. Our setting has applications where A has no eigenvalues or A is nota a self-adjoint operator bounded from above. See 4.1 and 4.2 respectively.

We also consider an observation operator $C \in \mathcal{L}(D(A), \mathcal{F})$ admissible for this semigroup, i.e, C is a continuous operator from $D(A)$ with the graph norm to another Hilbert space $\mathcal{F}$ and satisfies

ssibility}} $$\int_0^T \|Ce^{tA}x\|^2 dt \leq \mathrm{Adm}_T \|x\|^2, \qquad x \in D(A), \quad T > 0. \tag{4.2}$$

The admissibility constant $T \mapsto \mathrm{Adm}_T > 0$ is non-decreasing.

We say that (A, C, T) is observable at cost $\kappa_T > 0$ if

abilitykT}} $$\|e^{TA}x\|^2 \leq \kappa_T \int_0^T \|Ce^{tA}x\|^2 dt, \qquad x \in D(A). \tag{4.3}$$

Here, $\sqrt{\kappa_T}$ does not grow more than the semigroup as $T \to +\infty$, i.e, $T \mapsto \kappa_T$ is non-increasing when A is a non-positive self-adjoint operator.

We are interested in the asymptotic growth of κ_T as $T \to 0$ and think of κ_T as the cost of fast control.

Lemma 4.1. *If the approximate observability estimate*

$$f(t)\|e^{tA}x\|^2 - f(qt)\|x\|^2 \leq \int_0^t \|Ce^{\tau A}x\|^2 d\tau, \qquad x \in D(A), \quad t \in (0, T'], \tag{4.4}$$

holds with $f(t) \to 0$ as $t \to 0^+$, $q \in (0,1)$ and $T' > 0$, then $\kappa_T \leq 1f((1-q)T)$ for $T \in (0, T']$, i.e, the fast control cost does not grow more than the inverse of f.

4.2.2. *Relative observability on growth subspaces.* We assume that there is a non-decreasing familiy of semigroup invariant spaces $\mathcal{E}_\lambda \subset \mathcal{E}$, $\lambda > 0$, i.e, $e^{tA}\mathcal{E}_\lambda \subset \mathcal{E}_\lambda \subset \mathcal{E}_{\lambda'}$ for $t > 0$ and $\lambda' > \lambda$. This familty satisfies the growth condition

wthSpaces}} $$\|e^{tA}x\| \leq m_0 e^{-\lambda t}\|x\|, \quad x \perp \mathcal{E}_\lambda, \quad t \in (0, T_0), \quad \lambda > 0. \tag{4.5}$$

We call them growth spaces. We think them as spectral subspaces of A, i.e,

$$\sigma(A\,|_{\mathcal{E}_\lambda^\perp}) \subset \{z \in \sigma(A) \mid \Re z \leq -\lambda\},$$

and we think of (4.5) as a spectrally determined growth property.

We also assume that there is an admissible observation operator C_0 satisfying the bound relative to C on growth spaces with exponent $\alpha \in (0,1)$ and rate $a > 0$,

eOperator}}

$$\|C_0x\|^2 \le a_0 e^{2a\lambda^\alpha}\|Cx\|^2, \qquad x\in\mathcal{E}_\lambda, \quad \lambda>0. \tag{4.6}$$

We call C_0 a reference operator and the property (4.6) as observability on growth subspaces relatively to C_0. We think of C_0 as a simple operator with a good estimate of fast control like the identity operator.

4.2.3. *Main result.* When the reference operator C_0 satisfies the observability cost estimate with exponent $\beta>0$ and rate $b>0$,

$$\|e^{TA}x\|^2 \le b_0 e^{\frac{2b}{T^\beta}}\int_0^T \|C_0e^{tA}x\|^2dt, \quad x\in D(A), \quad T\in(0,T_0), \tag{4.7}$$

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we claim that C satisfies an observability cost estimate with exponent $\max\{\beta,\frac{\alpha}{1-\alpha}\}$,

{Theorem2.2}

Theorem 4.1. *Under the assumptions* (4.5), (4.6) *and* (4.7) *with* $\beta=\frac{\alpha}{1-\alpha}$, *the system* (A,C,T) *is observable at a cost* κ_T *such that* $2c=\limsup_{T\to0}T^\beta\ln\kappa_T<\infty$. *More precisely, this rate* c *is bounded in terms of an implicitly defined* $s>0$*:*

$$c\le c_*:=\left(\frac{b}{a}(\beta+1)\right)^{\frac{\beta+1}{\beta}}\frac{\beta^\beta}{s^{\frac{(\beta+1)^2}{\beta}}}, \quad s(s+\beta+1)^\beta=(\beta+1)\beta^{\frac{\beta^2}{\beta+1}}\frac{b^{\frac{1}{\beta+1}}}{a}. \tag{4.8}$$

Moreover, if the admissibility constant in (4.2) *satisfies* $\mathrm{Adm}_T\to0$ *as* $T\to0$, *then there exists* $T'>0$ *such that* $\kappa_T\le 4a_0b_0\exp\left(\frac{2c_*}{T^\beta}\right)$ *for* $T\in(0,T']$.

4.3. **Comments and consequences.**

4.3.1. *Admissibility.* Any $C\in(\mathcal{E},\mathcal{F})$ satisfies the admissibility condition 4.2 with $\mathrm{Adm}_T=T\|C\|^2$. It might be useful to circumvent the admissibility assumption

Lemma 4.2. *The conclusion* $2c=\limsup_{T\to0}T^\beta\ln\kappa_T\le 2c_*$ *of Theorem 4.1 is still valid if we replace the assumption that* $C\in\mathcal{L}(D(A),\mathcal{F})$ *satisfies the admissiblity condition* (4.2) *by the following time smoothing effect assumption*

$$\forall x\in\mathcal{E}\ \forall t>0,\ e^{tA}x\in D(A), \quad \limsup_{t\to0}t^\beta\ln\|Ae^{tA}\|=0. \tag{4.9}$$

4.3.2. *Controllability cost.* From the definition of κ_T in (4.3), we have $\kappa_T\le\|e^{(T-T')A}\|^2\kappa_{T'}$ for $T'<T$. This justifies why $\sqrt{\kappa_T}$ does not grow more than the semigroup as $T\to+\infty$.

Moreover, if $\kappa_t\le f(t)$ for $t\in(0,T']$ with g non-increasgint, then $\kappa_t\le Mg(t)$ for $t\in(0,T]$ with $M=\sup_{s\in(0,T-T')}\|e^{sA}\|^2<\infty$. This justifies that we restrict to some bounded intervales $(0,T']$ in the statement of our results.

The dual problem to the final-observability is the null-controllability of

$$\dot f(t)=A^*f(t)+Bu(t), \qquad f(0)=f_0\in\mathcal{E},\ t\ge0, \tag{4.10}$$

{{Eq18}}

with input $u\in L^2([0,T];\mathcal{F})$ and control operator $B=C^*\in\mathcal{L}(\mathcal{F},D(A^*))$. Since C satisfies the admissibility condition (4.2), B satisfies

$$\left\|\int_0^T e^{tA^*}Bu(t)\right\|^2\le K_T\int_0^T\|u(t)\|^2dt,$$

and the solution of (4.10) is

$$f(T)=e^{TA^*}f_0+\int_0^T e^{(T-t)A^*}Bu(t)dt.$$

More precisely, if (A,C,T) is observable at cost κ_T, then, for all f_0, there is a u such that $f(T)=0$ and $\int_0^T\|u(t)\|^2dt\le\kappa_T\|f_0\|^2$.

4.3.3. *Integrated observability estimate.* Lemma 2.1 can be seen as the discrete version of the following lemma which has been used with $f(t) = \exp(-c/t)$ when proving observability by some parabolic global Carleman estimate

Lemma 4.3. *If the integrated observability estimate*

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$$\int_0^T f(t)\|e^{tA}x\|^2 dt \leq \int_0^T \|Ce^{tA}x\|^2 dt, \quad x \in D(A), \quad T \in (0, T_0), \tag{4.11}$$

holds with $T_0 > 0$ and f is an increasing function such that $f(t) \to 0$ when $t \to 0^+$, then $\kappa_T \leq M_T/f(T)$ for $T \in (0, T_0)$ with $M_T = \sup_{t\in[0,T]} \|e^{tA}\|^2 \leq M_{T_0} < \infty$, i.e, the fast contorl cost does not grow more than the inverse of f. Conversely, if (4.3) holds for $T \in (0, T_0)$, then (4.11) holds for $T \in (0, T_0)$ with $f(T) = 1/(T_0\kappa_T)$.

4.3.4. *Approximate observability.* The following lemma clarifies the connection of Lemma 2.1 and the approximate controllability.

Lemma 4.4. *If (A, C, T) satisfy the approximate observability estimate,*

$$\|e^{TA}x\|^2 \leq \kappa \int_0^T \|Ce^{tA}x\|^2 dt + \varepsilon\|x\|^2, \quad x \in D(A), \tag{4.12}$$

for some positive κ and ε, then approximate null-controllability of (4.10) holds, i.e, for all f_0 there exists u such that $\|f(T)\|^2 \leq \varepsilon\|f_0\|^2$ and $\int_0^T \|u(t)\|^2 dt \leq \kappa\|f_0\|^2$.

4.3.5. *Normal semigroups and spectral spaces.* For a normal semigroup, the natural growth spaces are their spectral spaces. It has a spectral decomposition E, which is a projection-valued measure which commutes with any operator which commutes with A and defines spectral projections $E_\lambda = E(\{z \in \sigma(A) \mid \Re z > -\lambda\})$ and spectral spaces $\mathcal{E}_\lambda = E_\lambda\mathcal{E}$. Then (4.5) holds if the semigroup satisfies the spectral bound equal growth bound condition, i.e, it is eventually norm-continuous which happens when A is self-adjoint and bounded from above.

If there is an orthonormal basis $\{e_n\}$ of $\mathcal{E}$ such that $-Ae_n = \lambda_n e_n$, then the spectral spaces are just spanned by linear combinations of normalized eigenfunctions $\mathcal{E}_\lambda = \mathrm{Span}\{e_n\}_{\lambda_n<\lambda}$ and (4.6) becomes an estimate on sums of eigenfunctions of A.

When $A = -\Delta$ on $\mathcal{E} = L^2(\mathbb{R}^d)$, the spectral decomposition is the Fourier transform

$$\widehat{f(-\Delta)\phi}(\xi) = f(|\xi|)^2\hat{\phi}(\xi), \quad \phi \in L^2(\mathbb{R}^d),$$

thus $\phi \in \mathcal{E}_\lambda$ just means $\hat{\phi}(\xi) = 0$ for $|\xi|^2 > \lambda$, i.e, ϕ is the restriction to eh real axis of an entire function $\tilde{\phi}$ such that $|\tilde{\phi}(z)| \leq ce^{\sqrt{\lambda}|\Im z|}$ by the Paley-Wiener theorem.

4.3.6. *Original example.* For $A = \Delta$ the Dirichlet Laplacian on a compact smooth connected Riemannian manifold M, we use $\mathcal{E} = \mathcal{F} = L^2(M)$ and $\mathcal{E}_\lambda$ is the spectral space for a given level λ for A. C_0 is the identity operator and C is the multiplication by the characteristic function of an open subset $\Omega \neq \emptyset$ of $\overline{M}$.

4.4. **Applications.**

4.4.1. *Anomalous diffusions.* Let M be a smooth connected compelte $d-$ dimensional Riemannian manifold with metric g and boundary ∂M. When $\partial M \neq \emptyset$, M denotes the interior and $\overline{M} = M \cup \partial M$. Let Δ denote the Laplace-Beltrami oeprator on $L^2(M)$ with domaind $D(\Delta) = H_0^1(M) \cap H^2(M)$ defined by g.

Here, $\mathcal{E} = \mathcal{F} = L^2(M)$, $\mathcal{E}_\lambda$ are the spectral spaces and the reference operator C_0 is the identity operator and the observation operator C is the multiplication by $\mathbf{1}_\Omega$, where $\emptyset \neq \Omega \subset \overline{M}$. If M is not compact, assume that Ω is the exterior of a compact set K such that $K \cap \overline{\Omega} \cap \partial M = \emptyset$.

For $A = -(-\Delta)^\gamma$ and $\gamma > 1/2$, (4.6) holds with exponent $\alpha = \frac{1}{2\gamma}$. Applying Theorem 2.2, it improves the following result.

Theorem 4.2. *For all $\gamma > 1/2$, the anomalous diffusion:*

$$\partial_t \phi + (-\Delta)^\gamma \phi + \mathbf{1}_\Omega u, \quad \phi(0) = \phi_0 \in L^2(M),\ u \in L^2([0,T] \times M),$$

is null-controllable in any tume $T > 0$, Moreover, the cost κ_T satisfies $\limsup_{T\to 0} T^\beta \ln \kappa_T < \infty$ with $\beta = \frac{1}{2\gamma - 1}$.

4.4.2. *Structural damping.* Let A be a positive self-adjoint and boundedly invertible operator on another Hilbert space H. Let $D(A)$ be its domain with the norm $\zeta \mapsto \|A\zeta\|$. Let H_λ with $\lambda > 0$ denote the spectral spaces of A corresponding to $\{z \in \sigma(A) \mid \Re z < \lambda\}$. We consider two observation operators C_0 and C in $\mathcal{L}(D(A), \mathcal{F})$ satisfyinf the relative observability

$$\|C_0 z\|^2 \le d_0 e^{2d\lambda^\delta} \|Cz\|^2, \quad z \in H_\lambda, \quad \lambda > 0, \tag{4.13}$$ {{MainAssumpti

and a control operator $B = C^* \in \mathcal{L}(F, D(A)')$. To give a precise meaning to the solution of the structurally damped system

$$\begin{cases} z\ddot{e}ta(t) + \rho A^{2\gamma} \dot\zeta(t) + A^2 \zeta(t) = Bu(t), \\ \zeta(0) = \zeta_0 \in D(A), \\ \dot\zeta(0) = \zeta_1 \in H, \\ u \in L^2([0,T], \mathcal{F}), \end{cases} \tag{4.14}$$ {{StructuralDa

with structural dissipation power $\gamma \in (0,1)$, we write it as a first order system with state space $\mathcal{E} = D(A) \times H$,

$$A = \begin{pmatrix} 0 & I \\ -A^2 & -\rho A^{2\gamma} \end{pmatrix}, \quad D(A) = \{(z_0, z_1) \in \mathcal{E} \mid Az_0 + \rho A^{2\gamma - 1} z_1 \in D(A)\}.$$

It inherits from $-A$ the necessary and sufficient properties of Lumer-Phillips for generating a contraction semigroup.

The observation and control operators are the projection $C_0 : \mathcal{E} \to H$ defined by $C_0(z_0, z_1) = z_1$, $\tilde C = CC_0$ and B defined as before. We assume that $\tilde C$ is admissible for the semigroup generated by A. We entend the action of the orthogonal projection $\tilde H_\lambda$ on H_λ from H to $\mathcal{E}$ according to $H_\lambda(z_0, z_1) = (H_\lambda z_0, H_\lambda z_1)$. The growth condition 4.5 is proved by Miller for $\mathcal{E}_\lambda = H_\mu(\mathcal{E})$ with $\lambda = \min\{\frac{\rho}{2}, \frac{1}{\rho}\}\mu^{2\min\{\gamma, 1-\gamma\}}$. For this choice, the relative observability (4.6) holds with

$$\alpha = \frac{\delta}{2\min\{\gamma, 1-\gamma\}}, \qquad a = \frac{d}{\min\{\frac{\rho}{2}, \frac{1}{\rho}\}^\alpha}.$$

Applying theorem 2.2 and corollary, we get

Theorem 4.3. *Recall that δ and d are the exponent and rate in the main assumption 4.13. For all $\rho > 0$ and $\gamma \in (\delta/2, 1 - \delta/2)$, for all ζ_0, ζ_1, there is an input u such that the solution ζ of (4.14) satisfies $\zeta(T) = \dot\zeta(T) = 0$ and the cost estimate*

$$\int_0^T \|u(t)\|^2 dt \le b_0 \exp\left(\frac{2b}{T^\beta}\right) (\|A\zeta_0\|^2 + \|\zeta_1\|^2),\ \zeta_0 \in D(A),\ \zeta_1 \in H,$$

where T is small and

$$\beta = \left(\frac{2}{\delta}\min\{\gamma, 1-\gamma\} - 1\right)^{-1}, \qquad b > \frac{d^{\beta+1}}{\min\{\rho/2, 1/\rho\}^\beta} \frac{(\beta+1)^{\beta(\beta+1)}}{\beta^{\beta^2}}.$$

The main application of the previous is to the plate equation with square root damping and interior control in Ω with hinged boudnary conditions on a manifold M:

$$\begin{cases} \ddot\zeta - \rho\Delta\dot\zeta + \Delta^2\zeta = \mathbf{1}_\Omega u, & [0,T] \times M, \\ \zeta = \Delta\zeta = 0, & [0,T] \times \partial M, \\ \zeta(0) = \zeta_0 \in H^2(M) \cap H_0^1(M), \\ \dot\zeta(0) = \zeta_1 \in L^2(M), \\ u \in L^2([0,T] \times M). \end{cases} \tag{4.15}$$

Applying the previous theorem to $A = -\Delta$ with $\delta = \gamma = 1/2$, we get a better result than Mil06, Thm1, improving the value of β.

Under the geometrical optics condition from Brdos-Lebeau-Rauch, that the length L_Ω of the longest generalized geodesic in $\overline{M}$ which does not intersect Ω is not ∞.

4.4.3. *Diffusion in a potential well.* We consider $V(x) = |x|^{2k}$ with $x \in \mathbb{R}^d$ and $k \in \mathbb{N}$. Consider the Schrödinger operator $A = \Delta - V$ with domain $D(A) = \{\phi \in H^2(\mathbb{R}^d) \mid \int |V\phi|^2 < \infty\}$ is negative, self-adjoint and has compact resolvent. Denote

$$\Gamma = \{x \in \mathbb{R}^d \mid |x| > r_0,\ x/|x| \in \Omega_0\},$$

where $r_0 \geq 0$ and Ω_0 is an open subset of the unit sphere.

In this application we use $\mathcal{E} = \mathcal{F} = L^2(\mathbb{R}^d)$, the growth spaces are the spectral spaces, the reference operator C_0 is the identity operator and the observation operator C is the multiplication by $\mathbf{1}_\Gamma$.
In Mil08, (4.6) is achieved with exponent $\alpha = \frac{1}{2}(1 + \frac{1}{k})$. This allow us to use Theorem 2.2 and Lemma 3.5 to get

Theorem 4.4. *For all $k > 1$, the diffusion in the potential well $V(x) = |x|^{2k}$:*

$$\partial_t \phi - \Delta\phi + V\phi = \mathbf{1}_\Gamma u,\ \phi(0) = \phi_0 \in L^2(\mathbb{R}^d),\ u \in L^2([0,T] \times \mathbb{R}^d),$$

is null-controllable in any time $T > 0$. Moreover, the cost κ_T satisfies $\kappa = \limsup_{T\to 0} T^\beta \ln \kappa_T < \infty$ with $\beta = 1 + \frac{2}{k-1}$.

If there is a vector space of dimension 2 in $\mathbb{R}^d$ which does not intersect the closure $\overline{\Omega_0}$, then $\kappa \neq 0$.

4.5. **On the rate in the estimate on sums of eigenfunctions.**

Theorem 4.5. *Consider the Laplace-Beltrami operator Δ with Dirichlet obundary condition on a compact smooth connected Riemannian manifold M and an open subset $\Omega \neq \emptyset$ of $\overline{M}$. If the observability estimate*

$$\int_M |\phi(x)|^2 dx \leq c_0 e^{x\sqrt{\lambda}} \int_\Omega |\phi(x)|^2 dx,$$

holds for any $\lambda > 0$ and any sum of eigenfunctions

$$\phi = \sum_{\mu \in \sigma(-\Delta) \cap (0,\lambda)} \varphi_\mu, \qquad -\Delta\varphi_\mu = \mu\varphi_\mu,$$

then $c \geq \sup_{y\in M} \operatorname{dist}(y, \overline{\Omega})$.

Proof: The proof relies on an estimate of the heat kernel in small time.

By the following L^2 estimate, which is a consequence of the finite propagation for the wave equation and Kannai's formula,

$$e^{t\Delta} = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{+\infty} e^{-\frac{-s^2}{4t}} \cos(s\sqrt{-\Delta}) ds,$$

we get using C (resp. C') as the multiplication by $\mathbf{1}_\Omega$ (resp. $\mathbf{1}_{\Omega'}$), that

{CEstimate}} $$\|Ce^{t\Delta}C'\|_{\mathcal{L}(L^2)} \leq \frac{1}{\sqrt{4\pi t}} \int_{|s|\geq\rho} e^{-\frac{-s^2}{4t}} ds \leq e^{-\frac{\rho^2}{4t}}. \tag{4.16}$$

Here, $\rho = \operatorname{dist}(\Omega, \Omega')$.

Here we use functional calculus. Denote by $\mathbf{1}_{-\Delta<\lambda^2}$ the projection on the space spanned by the eigenfunctions of $-\Delta$ with eigenvalues less tan λ^2. Fix an eigenvalue μ^2 of $-\Delta$ and φ_μ the corresponding normalized eigenfunction. The unique continuation property for elliptic operators implies that $Y = \{y \in M \setminus \overline{\Omega} \mid \varphi_\mu(y) \neq 0\}$ is an open dense set in $M \setminus \overline{\Omega}$ so that the supremum on the previous theorem can be taken over $y \in Y$

instead of $y \in M$. Since eigenfunctions are continuos, for any $y \in Y$ we may find a small ball Ω' with center y such that

$$\left| \int_{\Omega'} \varphi_\mu(x) dx \right| > 0,$$

and $\mathrm{dist}(\Omega, \Omega')$ is arbitrarily close to $\mathrm{dist}(y, \overline{\Omega})$. With these notation, we asume that

$$\|\phi\| \le a_0 e^{a\lambda} \|C\phi\|, \quad \phi \in \mathbf{1}_{-\Delta<\lambda^2}(L^2(M)), \quad \lambda > 0, \tag{4.17}$$ {{32}}

and prove $2a \ge \rho = \mathrm{dist}(\Omega, \Omega')$, for any opern subset $\Omega' \neq \emptyset$ with characteristic function χ satisfying

$$|\langle \chi, \varphi_\mu \rangle| > 0. \tag{4.18}$$ {{33}}

For any $\lambda > 0$, consider $\phi_\lambda = e^{\frac{\rho}{2\lambda}\Delta}\chi$ and $\phi = \mathbf{1}_{-\Delta<\lambda^2}\phi_\lambda$. In (4.16), C' is the multiplication by χ so that $C'\chi = \chi^2 = \chi$ and (4.16) yields

$$\|C\phi_\lambda\| \le e^{-\rho\lambda/2}\|\chi\|.$$

Since $\phi_\lambda - \phi = \mathbf{1}_{-\Delta\ge\lambda^2} e^{\frac{\rho}{2\lambda}\Delta}\chi$, the spectral theorem yields

$$\|C(\phi_\lambda - \phi)\| \le \|\phi_\lambda - \phi\| \le e^{-\frac{\rho\lambda}{2}}\|\chi\|$$

Adding these inequalities yields $\|C\phi\| \le 2e^{-\frac{\rho\lambda}{2}}\|\chi\|$ and since $\|\varphi_\mu\| = 1$ and $-\Delta\varphi_\mu = \mu^2\varphi_\mu$, we also have $\|\phi\| \ge |\langle \phi, \varphi_\mu \rangle| + |\langle \chi, e^{\frac{\rho}{2\lambda}\Delta}\varphi_\mu \rangle = e^{-\frac{\rho\mu^2}{2\lambda}}|\langle \chi, \varphi_\mu \rangle|$.

Using these inequalities in (4.17), using (4.18) yields

$$0 < e^{-\frac{\rho\mu}{2}}|\langle \chi, \varphi_\mu \rangle| \le 2a_0 e^{(a-\rho/2)\lambda}\|\chi\|, \qquad \lambda \ge \mu.$$

We finally see that $a < \rho/2$ is forbidden by taking $\lambda \to \infty$. □